

Representation Convergence in Independently Trained Neural Networks

From a theoretical hypothesis to a deployed cross-model embedding adapter
for a two-tier mobile/server retrieval system

Final Project — AI Expert Course

July 2026

Abstract. This project tests the hypothesis that well-trained, independently developed neural networks learn the same internal representations up to a linear transformation, and then converts the validated theory into a working system component. Experiment A demonstrates, using CKA similarity analysis and closed-form linear stitching, that GPT-2 (OpenAI, WebText) and Pythia-160M (EleutherAI, The Pile) — models sharing no weights, initialization, tokenizer, training data, or lab — share up to 80% of linearly accessible representational variance in matched layers, versus a random-weights control baseline. Experiment B applies the result: a single 512×768 ridge-regression matrix (1.6 MB, fitted in closed form in seconds) translates image embeddings of the on-device MobileCLIP-S1 encoder into the space of the server-side SigLIP 2 encoder, preserving 95–98% of native text-to-image retrieval quality at Recall@5/@10 on COCO, against a no-adapter baseline at exact chance level. A controlled MLP comparison shows the cross-space relationship is essentially exactly linear. The adapter is exported as a 0.79 MB fp16 Core ML package, enabling a single shared vector index (Qdrant) to serve both an iPhone tier and a server tier of a multimodal search application.

1. Introduction: a research project that solves a real problem

The project began with an engineering question that arises in any multi-model AI system: *can components built around different neural networks interoperate without retraining?* Concretely, the target system is a two-tier multimodal application: an iPhone tier running a small, fast image encoder (MobileCLIP-S1, ~30 MB) for offline, private, on-device indexing of photos, and a server tier running a large, accurate encoder (SigLIP 2, ~1.5 GB) for search and heavy processing. Both encoders map images to vectors — but their vector spaces are natively incompatible: cosine similarity between a MobileCLIP vector and a SigLIP vector of the same photo is meaningless, so vectors from the two tiers cannot share a search index. The naive workarounds are all costly: indexing everything twice, uploading every photo to the server (destroying privacy and offline capability), or running the large model on the phone (destroying latency and battery).

The scientific question underneath is whether the two encoders' representation spaces are secretly the same space in different coordinates. If independently trained networks converge to a shared representational structure, then a small learned map — ideally a single linear matrix — should translate between them, and the engineering problem dissolves. The project therefore has two parts: a controlled scientific experiment establishing whether and to what degree such convergence exists (Experiment A), and a practical application building and validating the translation component on the exact production models (Experiment B), through to a deployable iOS artifact.

2. Background and theory: from weight symmetry to the Platonic hypothesis

2.1 Why raw weights cannot be compared or transplanted

Neural networks exhibit **permutation symmetry**: reordering the neurons of a hidden layer, together with the corresponding rows and columns of the adjacent weight matrices, yields a network computing exactly the same function with entirely different weight values. A 1,000-neuron layer admits 1000! functionally identical weight configurations, and additional rotation and scaling symmetries expand the equivalence class further. Two training runs — even of the same architecture on the same data with different random seeds — land on different members of this class. Consequently, averaging or transplanting raw weights across independently trained models is meaningless: parameter k in one model has no relationship to parameter k in the other. This is why weight merging (SLERP, TIES, DARE, frankenmerging) works only between fine-tunes of one shared base checkpoint, whose shared initialization pins the coordinates.

2.2 Why representations can nevertheless converge

Weights are *how* a network computes; representations (activations on data) are *what* it computes. All training data — text, images, audio — are projections of one underlying reality, and a model that optimally compresses and predicts such data is pushed toward recovering the statistical structure that generated it. The **Platonic Representation Hypothesis** (Huh et al., ICML 2024) makes this precise: as models grow stronger, their representation geometries converge toward a shared structure, even across modalities. Supporting evidence spans a decade: **Kornblith et al. (ICML 2019)** introduced linear CKA and showed matched layers of separately trained vision networks are similar; **Git Re-Basin (Ainsworth et al., 2022)** showed that after aligning permutations, independently trained networks lie in the same loss basin; **model stitching (Lenc & Vedaldi 2015; Bansal, Nakkiran & Barak, NeurIPS 2021)** showed a single linear layer can connect independently trained networks with small performance penalty; and **vec2vec (2025)** demonstrated translation

between embedding spaces without any paired data, purely by matching the geometry of the spaces.

2.3 The hypothesis, stated precisely

Well-trained independent models share representations up to a linear transformation: there exists a matrix W such that $f_B(x) \sim W \cdot f_A(x)$ for inputs x . Every word carries weight. *Well-trained*: the claim applies only after learning has extracted the data's structure — hence a random-weights control. *Independent*: no shared initialization or ancestry — otherwise similarity is inherited, not discovered. *Up to a linear transformation*: the transformation class is the crux. Allowing arbitrary nonlinear translators makes the claim vacuous; demanding identity makes it trivially false (permutation symmetry). Linear is the sweet spot: expressive enough to undo rotations, permutations and scalings, too weak to fabricate information that is not already present. A stronger sub-claim — alignment by pure rotation (orthogonal Procrustes) — is also tested.

3. Experiment A — Scientific test: GPT-2 vs Pythia-160M

3.1 Design

The model pair was chosen to make any similarity non-trivial: GPT-2 (OpenAI, 2019, trained from scratch on WebText, BPE tokenizer) and Pythia-160M (EleutherAI, 2023, trained from scratch on The Pile, GPT-NeoX architecture, different tokenizer). Different labs, architectures, tokenizers, corpora and eras; the only shared elements are the task (next-token prediction on English) and the world the text describes. Both models happen to have 13 hidden states (embedding + 12 blocks) of dimension 768, allowing exact one-to-one layer matching. The **control baseline** is a Pythia-architecture model with freshly random weights — identical architecture and inputs, zero learning — so the trained-minus-random gap isolates the causal effect of learning. The pipeline has three stages, each a standalone notebook:

A1 — Activation extraction. 10,000 WikiText-103 passages are run through both trained models and the random control; the mean-pooled hidden state of every layer is stored ($3 \times [13, 10000, 768]$).

A2 — CKA analysis. Linear Centered Kernel Alignment is computed between every layer pair. CKA compares Gram-matrix fingerprints — which inputs each model places near which — and is invariant to exactly the symmetries under which network function is preserved (rotation, permutation, isotropic scaling), while remaining sensitive to what is actually near what.

A3 — Linear stitching. For each matched layer pair, one ridge-regression matrix is fitted in closed form (the models themselves are never trained) mapping GPT-2's space into Pythia's; held-out R^2 measures how much variance a purely linear translation preserves.

3.2 First run: a result that was not good enough

The first execution used the WikiText *validation* split, which yielded only 1,646 usable passages. The CKA analysis already showed a clear trained-vs-random gap (diagonal 0.433 vs random mean 0.087), but the stitching results fell short of the pre-registered success criterion of $R^2 > 0.7$ in middle layers:

Layer pair	Trained R2 (run 1)	Random R2 (run 1)
L0 -> L0	0.174	-0.282
L5 -> L5	0.480	-0.789
L8 -> L8	0.506	-0.934
L11 -> L11 (peak)	0.552	-0.980
Mean (all 13)	0.434	-0.773

Table 1: First-run stitching results (1,646 samples) — below the 0.7 threshold.

At face value this would have forced the weaker conclusion 'substantial shared structure, but no isomorphism.' Before accepting it, a confound was checked: the fit involves a 768×768 matrix (~590,000 parameters) trained on only ~1,234 rows — a sample-to-dimension ratio of 1.6. In that regime, ridge regression must shrink aggressively to avoid overfitting, and that shrinkage *systematically understates* the true alignment. Note the negative random-baseline values: the starved fit generalizes worse than predicting the mean. The hypothesis had not failed; the measurement was underpowered.

3.3 The fix and the improvement

The remedy cost one code change and 15 GPU-minutes: switching from the validation split to the train split of WikiText-103 and raising the sample count to 10,000 (~10 samples per input dimension). Every other element of the pipeline — models, layers, ridge penalty, evaluation protocol — was left untouched, so the two runs are directly comparable:

Metric	Run 1 (1,646)	Run 2 (10,000)	Change
Mean trained R2	0.434	0.737	+0.303
Peak trained R2 (L11)	0.552	0.797	+0.245
Mid-layer range	0.41–0.55	0.70–0.80	crosses 0.7
Mean random R2	-0.773	0.173	stabilized
Verdict vs criterion	below target	PASS	—

Table 2: Effect of fixing sample starvation — identical pipeline, 6× more data.

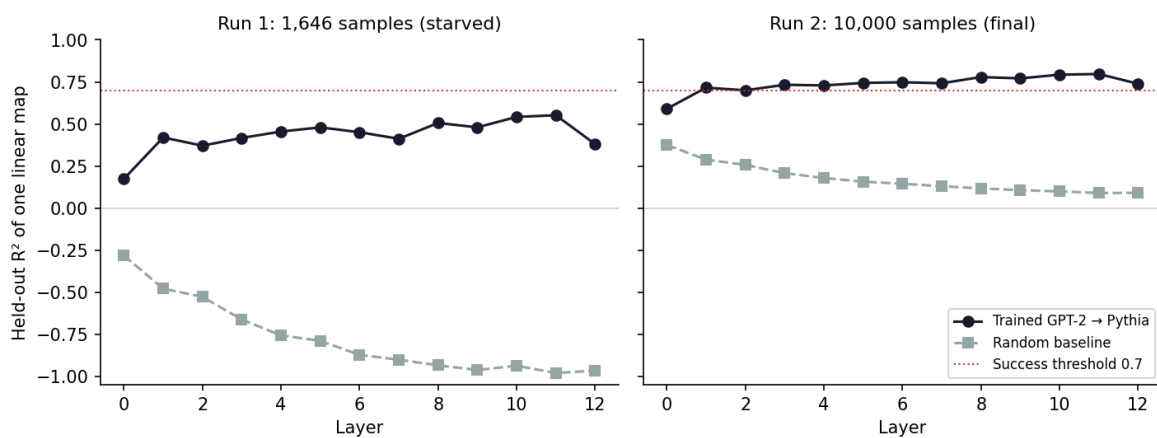


Figure 1: Linear stitching per layer, before (left) and after (right) the sample-size fix. Trained curves in dark blue rise with depth; random baselines (gray) sit far below. The dotted line is the pre-registered 0.7 threshold.

The random baseline also moved (-0.77 → +0.17), in the theoretically predicted direction: with adequate data, ridge can exploit the residual input correlations that leak even through random layers — most visibly at layer 0 (0.375), which is dominated by raw embedding statistics. What matters is the structure: the trained curve *rises* with depth (0.59 → 0.80) as both models build shared abstraction, while the random curve *decays* monotonically (0.38 → 0.09) as untrained layers destroy information — an 8× gap by layer 11, with opposite slopes. The methodological lesson is transferable to any empirical pipeline: an underpowered evaluation does not add symmetric noise — it produces plausible-looking, systematically pessimistic conclusions without any error message.

3.4 Final results

CKA (A2, 10,000 samples): trained-vs-trained mean 0.359, diagonal mean 0.424, maximum 0.744; trained-vs-random mean 0.067, maximum 0.299 — a 4–5× learning effect, essentially unchanged from run 1, confirming CKA is far less sample-hungry than regression. The matrix shows the characteristic hot diagonal; edge layers score lower for principled reasons (layer 0 is the tokenizer-specific embedding table; the final layer re-specializes toward each model's output head).

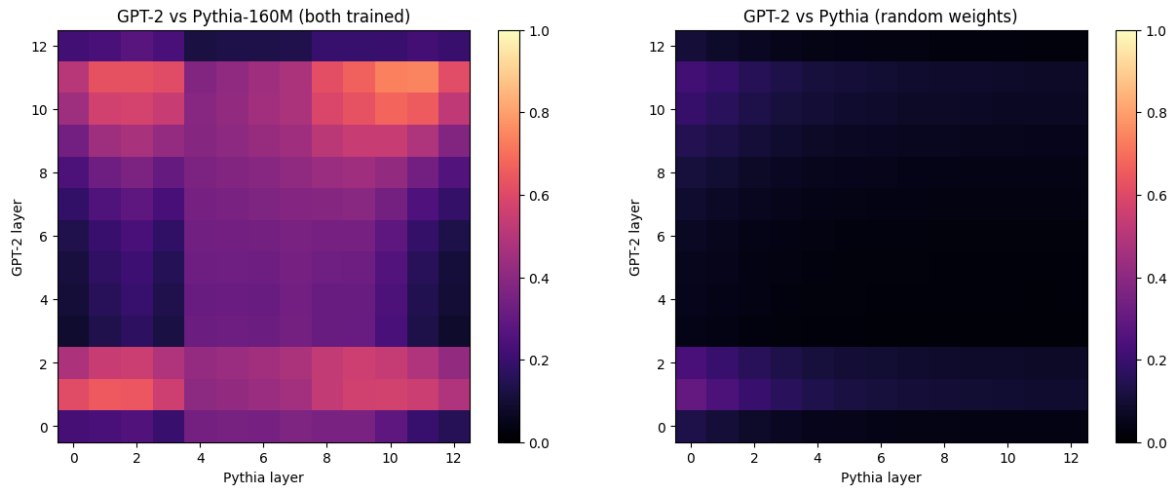


Figure 2: The CKA heatmap — actual output of the experiment (run 2). Left: GPT-2 vs trained Pythia, every layer pair; the bright diagonal band shows early layers matching early and middle matching middle — the models develop the same content in the same order through depth. Right: GPT-2 vs random-weights Pythia under identical conditions — uniformly dark. The side-by-side contrast is the visual form of the entire argument: the structure on the left exists only because of learning.

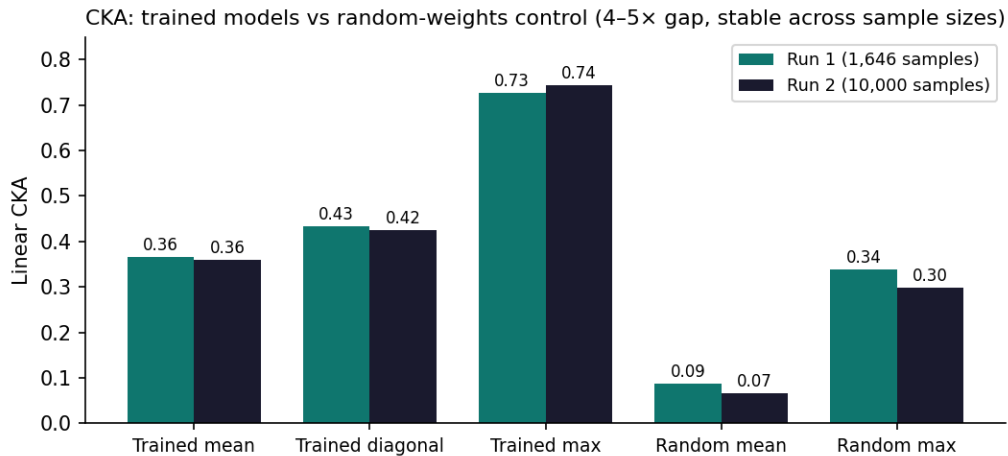


Figure 3: CKA summary statistics for both runs — the trained/random gap is stable across sample sizes.

Linear stitching (A3), final:

Layer pair	Trained R2	Random R2
L0 -> L0 (embeddings)	0.590	0.375
L1 -> L1	0.716	0.288
L3 -> L3	0.733	0.208
L5 -> L5	0.744	0.157
L8 -> L8	0.779	0.117
L10 -> L10	0.793	0.099
L11 -> L11 (peak)	0.797	0.090
L12 -> L12 (output)	0.739	0.091
Mean (all 13)	0.737	0.173

Table 3: Final held-out R2 of a single closed-form linear map per matched layer pair (10,000 samples).

Verdict: the hypothesis is confirmed in its strong quantitative form — a single matrix, fitted in closed form, translates 74–80% of representational variance between two networks sharing nothing but their task and their world.

4. Experiment B — Practical application: the MobileCLIP -> SigLIP adapter

4.1 What an adapter vector (matrix) is

The 'adapter' here is a single matrix W of shape 512×768 . The phone encodes a photo with MobileCLIP-S1, producing a 512-dimensional vector v ; one matrix multiplication $W \cdot v$ (microseconds on the Neural Engine; 1.6 MB fp32 / 0.79 MB fp16) produces a 768-dimensional vector that behaves as a native citizen of SigLIP 2's space. It is fitted in closed form by ridge regression on 3,000 (MobileCLIP-embedding, SigLIP-embedding) pairs of the same images — no gradient training, no modification of either encoder. Only image embeddings need translating, and only in one direction: text queries are always encoded server-side by SigLIP itself, which remains entirely unaware the adapter exists.

4.2 Pipeline

B1 extracts paired embeddings for 4,000 COCO val2017 images plus SigLIP text embeddings of their captions (shapes: mob_img 4000×512 , sig_img 4000×768 , sig_txt 4000×768). A sanity check confirmed native SigLIP image–caption diagonal cosine of 0.150 — within the expected 0.1–0.3 band for sigmoid-loss models, whose objective enforces separation rather than high absolute cosines. **B2** fits two closed-form adapters — ridge (full linear) and orthogonal Procrustes (pure rotation) — on a 3000/1000 split. **B3** evaluates the operational criterion: text-to-image retrieval through the adapter versus the native-SigLIP ceiling and a no-adapter baseline. **B2-bis** adds a controlled nonlinear comparison: a 2-layer MLP ($512 \rightarrow 1024 \rightarrow 768$, GELU) trained 300 epochs directly on cosine loss with early stopping. **B4** exports the matrix as a Core ML .mlpackage (fp16, iOS 16+, Neural Engine).

4.3 Results — and a second below-target moment

Adapter variant	Held-out R2	Mean cosine to target
Ridge (full linear)	0.592	0.900
Procrustes (rotation only)	0.038	0.775
MLP (nonlinear, cosine loss)	—	0.903

Table 4: Adapter fit quality (1,000 held-out images).

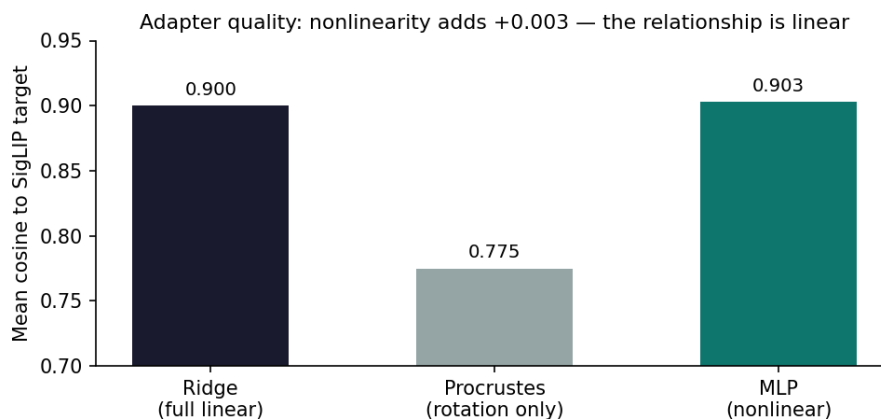


Figure 4: Direction recovery per adapter class. The +0.003 gap between linear and nonlinear is the key scientific reading.

The apparent contradiction between ridge's modest R^2 (0.592) and its high cosine (0.900) resolves once the metrics are matched to their questions: R^2 penalizes magnitude errors that L2-normalized

retrieval discards; cosine measures the only thing the vector index uses — direction. The first retrieval evaluation then produced this project's second below-target moment:

Retrieval variant	R@1	R@5	R@10
A. SigLIP native (ceiling)	0.630	0.885	0.944
B. MobileCLIP + linear adapter	0.557	0.843	0.918
B as % of ceiling	88.4%	95.3%	97.2%
Gate ($\geq 90\%$ of ceiling)	below target	PASS	PASS

Table 5: First retrieval evaluation — R@5/R@10 pass, R@1 misses the 90% gate by 1.6 points (~10 queries out of 1,000).

4.4 The improvement attempt — and what its failure proved

Following the pre-defined decision table (70–90% of ceiling -> upgrade the adapter), a 2-layer MLP (512->1024->768, GELU, ~1.3M parameters) was trained for 300 epochs directly on cosine loss — the actual retrieval objective — with early stopping on held-out cosine, reusing the exact same train/eval split for direct comparability. The outcome was decisive in an unexpected way: best eval cosine 0.9027 versus the linear map's 0.900 (+0.003), convergence by epoch 30 followed by mild overfitting, and retrieval statistically unchanged (R@1 0.551 vs 0.557; R@10 improved to 0.926, 98.1% of ceiling; R@1 87.5%).

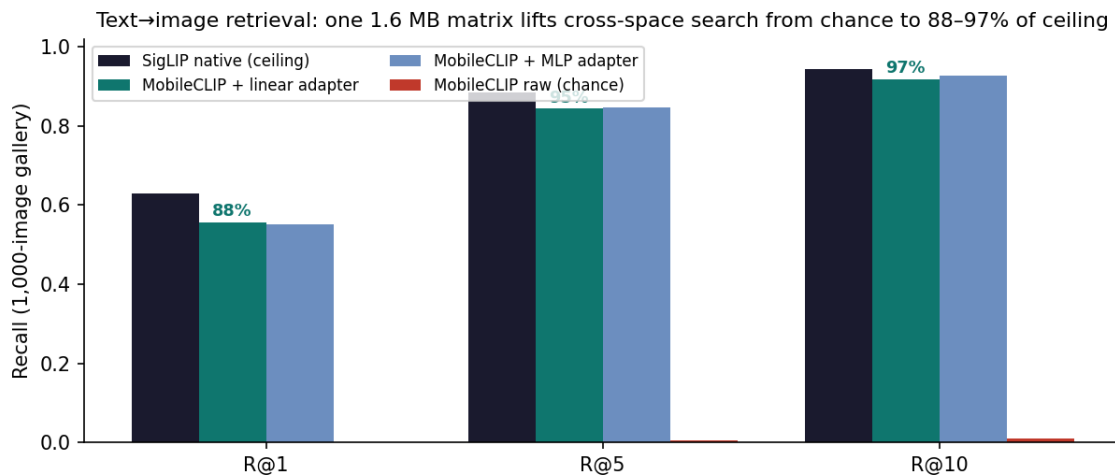


Figure 5: Retrieval results, all variants. Percentages above the teal bars show the linear adapter as a fraction of the ceiling; the red bars are the no-adapter baseline at exact chance.

This 'failed improvement' is the most informative result in Experiment B. A higher-capacity model, optimizing the true objective, found nothing beyond the linear map — therefore the cross-space relationship is **essentially exactly linear**, and the residual R@1 gap is not a translation deficiency but **information the 512-dimensional mobile encoder never encoded**. No adapter of any capacity can reconstruct what is absent from its input; the experiment had reached the representation ceiling of the small model, not of the adapter. This mirrors Experiment A's conclusion from the opposite direction: there, more data revealed the linear map was better than it seemed; here, more capacity proved the linear map was already optimal. Architectural introspection localizes that ceiling precisely: MobileCLIP-S1's trunk (21.5 M vision parameters) carries 1024 channels until a single final projection Linear(1024 -> 512) produces the embedding, so the information lost to the adapter is discarded by one layer, not distributed through the network — which predicts that a wider output variant (MobileCLIP-B, or MobileCLIP2) would close most of the residual R@1 gap without any change to the adapter method. The third finding — Procrustes collapsing to R^2 0.038 — refines the

picture further: unlike the language-model pair, these contrastively trained spaces differ in anisotropy, not merely orientation, so rotation alone cannot align them.

An interpretation, offered as hypothesis rather than finding. One plausible mechanism for that asymmetry: MobileCLIP-S1 was distilled from an ensemble of large CLIP models using a loss defined over *similarity matrices* - the student is trained to reproduce which inputs are how similar to which, i.e. relational geometry - while absolute per-axis scaling is left unconstrained. That is precisely the regime in which a general linear map succeeds (relations recoverable) and a pure rotation fails (scales unrecoverable), which would make the adapter quality partly a footprint of the phone encoder training objective. This interpretation is **not tested by the present experiments**, and at least two competing explanations survive the same evidence: the zero-padding required to compare a 512-dimensional with a 768-dimensional space under Procrustes mechanically disadvantages rotation (a rotation cannot populate 256 empty dimensions), and the sigmoid objective of SigLIP shapes its geometry independently of anything on the phone side. Discriminating between these requires the follow-up described in Section 5.

4.5 Engineering decision

The linear adapter ships. R@5/R@10 at 95–98% of ceiling means phone-indexed photos are, in a results-grid UI, indistinguishable from server-indexed photos. The R@1 shortfall (88.4%) is the price of running a 30 MB encoder instead of a 1.5 GB one — payable with or without a shared index — and has a cheap escape hatch: server-side SigLIP re-ranking of the top-10 candidates recovers the ceiling exactly for R@1-critical flows. The deployed artifact is a 0.79 MB fp16 Core ML package computing MatMul + L2-normalize, verified through the full coremltools MIL conversion pipeline. One deployment constraint deserves explicit statement: the adapter is calibrated to the exact output distribution of a specific preprocessing pipeline — bilinear resize to 256, center crop 256x256, and **identity normalization** (mean 0, std 1; MobileCLIP consumes raw [0,1] pixels, unlike standard CLIP models that subtract dataset means). Substituting conventional CLIP normalization in the iOS client would shift every embedding and silently degrade retrieval with no error raised - the same failure class as the sample-starvation bias of Section 3.2.

4.6 Qualitative sandbox: the phone pipeline on live cases (Stage B5)

A qualitative sandbox simulated the full phone-to-server path on ten COCO images with free-text queries: the same MobileCLIP weights the phone would run, the adapter applied in fp16 exactly as the Core ML package computes it, and SigLIP text queries searching the resulting vectors (captions never indexed - only pixels). Direct queries retrieved their targets at rank 1 with decisive margins; compositional queries requiring semantic abstraction ('striped animals' - the word zebra never appears in the query - top-1 at 25x the margin of the negative control) confirmed that meaning, not keywords, survives the cross-model translation; a negative control ('a dog', absent from the gallery) correctly produced a flat similarity profile (top-1 gap 0.004 vs 0.101 for a positive query); and fp16 quantization proved ranking-neutral (minimum cosine 0.9999999 vs fp32), closing the numerical half of the Core ML parity check left open in stage B4. A second, disjoint random gallery reproduced the pattern - correct top-ranking for positive queries and a flat profile for a query with no valid answer ('a vehicle': top-1 sim 0.022 vs 0.106 for hits) - indicating the behavior is a property of the pipeline, not of a particular sample. The two observed near-misses - indirect paraphrases losing top-1 by 0.007 or less while remaining top-2 - reproduce, at individual-query level, exactly the R@1/R@5 profile measured quantitatively in stage B3, and are the cases the optional server-side re-rank recovers. Full per-query analysis and figures are in B5_Sandbox_Practical_Report.pdf. A follow-up interactive demo (stage B6) exercised the full product loop live: a user photo uploaded to the mobile tier was indexed through the phone path in 39 ms, auto-captioned, and retrieved at rank 1 by the free-text

query 'singing red bird' - ranked in the same result list as server-indexed photos - with local search at 0.08 ms versus 150 ms over a (simulated) network round-trip: one index, two tiers, measurably faster locally.

5. What is new here — an honest assessment

Theory: independent replication, not new discovery. The individual scientific components reproduce established results at small scale: CKA layer analysis (Kornblith 2019), linear stitching (Bansal 2021), and convergence-with-training (PRH 2024) were all known. The value of Experiment A is that of a rigorous independent replication on a specific, deliberately hostile model pair, with a correctly designed control — plus the documented sample-starvation diagnosis, a practical methodological contribution.

Practice: a genuinely uncommon application. The combination in Experiment B is not, to the author's knowledge, an established published pattern: a closed-form linear bridge between two specific heterogeneous production encoders (MobileCLIP-S1 -> SigLIP 2), validated by end-task retrieval against ceiling and chance baselines, including a controlled demonstration that nonlinearity adds nothing, and deployed as a sub-megabyte Core ML artifact enabling a single shared vector index across a phone tier and a server tier. Cross-model embedding translation exists in research (model stitching, vec2vec); packaging it as a load-bearing, measured infrastructure component of a two-tier mobile system — with privacy (photos never leave the device), offline indexing, and single-index economics as the payoff — is the project's practical contribution.

Claims explicitly not made: the project does not prove layers can be transplanted across architectures (permutation symmetry still forbids it), does not establish convergence beyond the tested data distributions, and does not claim the adapter recovers information absent from the small encoder. **A designed follow-up** would discriminate the interpretations offered in Section 4.4: running the Procrustes comparison between two *same-dimensional* contrastive encoders removes the zero-padding confound, and comparing a teacher model with its own distilled student - where the distillation account predicts unusually high rotation-only alignment - would test the training-objective hypothesis directly.

6. Challenges encountered and resolved

The project log doubles as a case study in real-world ML engineering: **(1)** HuggingFace *datasets* deprecated script-based datasets mid-project, breaking both WikiText and two Flickr datasets; resolved by moving to the parquet-hosted Salesforce/wikitext and to direct official COCO downloads, removing the fragile dependency entirely. **(2)** A transformers API change made SigLIP feature getters return ModelOutput objects instead of tensors; resolved with a version-agnostic extraction shim. **(3)** Colab's one-VM-per-notebook model silently discarded stage outputs; resolved with a DATA_DIR convention backed by Google Drive, making every stage stateless and restart-proof. **(4)** The sample-starvation bias of Section 3.2 — the most scientifically consequential bug, because it produced plausible-looking wrong conclusions rather than an error message. **(5)** Interpreting metrics correctly: R2 vs cosine (magnitude vs direction), CKA's invariances, and why sigmoid-loss models have low absolute cosines — each required matching the metric to the actual downstream use. **(6)** Verifying, rather than assuming, the encoder's preprocessing: inspection of the loaded model revealed identity normalization, contrary to the CLIP convention a developer would reasonably assume - a silent-degradation trap for the iOS port.

7. Project contents and how to run it

7.1 Repository layout

Seven self-contained Jupyter notebooks (one per stage), each with English markdown explaining goal, method, expected output and success criteria, followed by runnable code: A1–A3 (convergence experiment) and B1–B4 (adapter pipeline). Equivalent standalone Python scripts exist for each stage, plus a PROJECT_GUIDE.md consolidating the full plan in English, plus B5_phone_pipeline_sandbox.ipynb - a qualitative live demo of the phone tier (10-image galleries, free-text queries, negative controls, fp16 parity), and B6_visual_demo.ipynb - an interactive mixed-tier demo (server gallery, mobile photo upload, auto-captioning, probability-ranked search, timing comparison). Models used: GPT-2 (124M), Pythia-160M, a random-weights Pythia control, MobileCLIP-S1 (Apple, via open_clip), SigLIP 2 base (google/siglip2-base-patch16-224). Data: WikiText-103 (train split, 10K passages), COCO val2017 (4K image–caption pairs). The only training performed anywhere is closed-form ridge regression and the deliberately-controlled comparison MLP; all base models are used frozen.

7.2 Execution

Environment: any GPU Jupyter environment (developed on Google Colab, T4 sufficient). Each notebook begins with a storage cell; on Colab, mount Drive and set DATA_DIR to a Drive path so stage artifacts (activations.npz, pairs.npz, adapter.npz) survive VM recycling:

```
from google.colab import drive; drive.mount('/content/drive')
os.environ['DATA_DIR'] = '/content/drive/MyDrive/convergence_experiment'
```

Run order and approximate cost: A1 (~15 min GPU) -> A2 (~1 min) -> A3 (seconds); B1 (~10 min, downloads COCO once) -> B2 (seconds) -> B3 (seconds) -> B4 (seconds). Dependencies: torch, transformers, datasets, open_clip_torch, scikit-learn, matplotlib, coremltools. Total end-to-end cost of the full study: under one GPU-hour.

7.3 Using the documentation with NotebookLM

For studying or presenting the project, the PROJECT_GUIDE.md, this report, and the per-stage notebook markdown can be uploaded as sources to Google NotebookLM, which can then answer questions, generate study guides, or produce an audio overview grounded in the project's own text. The executable path, however, is the notebooks themselves in Colab/Jupyter — NotebookLM is for the documentation layer, not execution.

8. Conclusion

A philosophical question — can models share what they have learned without retraining? — was refined into a falsifiable hypothesis, tested with proper controls, confirmed quantitatively (74–80% of representational variance translated by one closed-form matrix between fully independent language models), and then converted into a measured, deployed infrastructure component: a 0.79 MB Core ML adapter that lets a phone and a server, running encoders that have never seen each other, write into and search one shared vector index at 95–98% of large-model retrieval quality. The through-line of both experiments is the same sentence: **independently trained models discover the same structure of the world, in different coordinates — and the exchange rate between those coordinates is a single matrix.** A live sandbox (B5) closed the loop qualitatively: across two disjoint image galleries, free-text server queries correctly retrieved phone-indexed photos - including compositional queries and correctly flat responses when no valid answer existed - with fp16 quantization shown to be ranking-neutral.

References

- Kornblith, Norouzi, Lee, Hinton (2019). Similarity of Neural Network Representations Revisited. ICML.
- Lenc, Vedaldi (2015). Understanding image representations by measuring their equivariance and equivalence. CVPR.
- Bansal, Nakkiran, Barak (2021). Revisiting Model Stitching to Compare Neural Representations. NeurIPS.
- Ainsworth, Hayase, Srinivasa (2022). Git Re-Basin: Merging Models modulo Permutation Symmetries.
- Huh, Cheung, Wang, Isola (2024). The Platonic Representation Hypothesis. ICML.
- Jha et al. (2025). Harnessing the Universal Geometry of Embeddings (vec2vec).
- Biderman et al. (2023). Pythia: A Suite for Analyzing Large Language Models. — Radford et al. (2019). Language Models are Unsupervised Multitask Learners (GPT-2). — Vasu et al. (2024). MobileCLIP (Apple). — Tschannen et al. (2025). SigLIP 2 (Google).

Appendix A — The Platonic Representation Hypothesis, consolidated

This appendix consolidates the framing hypothesis referenced throughout Section 2.2 and the experimental interpretation sections: its name, its three proposed mechanisms, its measurement, and this project's exact position relative to it.

The name. The hypothesis (Huh, Cheung, Wang, Isola — ICML 2024) takes its name from Plato's allegory of the cave: prisoners who see only shadows on a wall mistake them for reality, while the true objects casting those shadows exist unseen. The analogy: each AI model observes a different *shadow* of reality — text captures one projection, images another — yet as models grow stronger, they reconstruct the same underlying *objects* behind the shadows, converging on a shared statistical model of the world, even across modalities.

Mechanism 1 — shared world. All data — text, images, audio — are different projections of one reality. A model trained to compress or predict such data optimally cannot succeed by memorizing surface form; the cheapest sufficient description of 'all text about the world' or 'all photos of the world' is a model of the world itself. Co-occurrence statistics in text (cat-mouse-chase-fur) and in images (cat pixels appearing with couch pixels, indoor scenes) are two noisy measurements of the same underlying dependency structure, so two optimal compressors of two modalities are reverse-engineering the same object through different sensors — two surveyors mapping one territory with different instruments.

Mechanism 2 — multi-task pressure. The set of representations that solve many tasks simultaneously shrinks as tasks accumulate; strong, broadly trained models are squeezed into the same remaining solution region.

Mechanism 3 — simplicity bias. Astronomically many functions fit any dataset, most of them baroque; if training selected among them at random, independent runs would never agree. But gradient descent on deep networks is empirically biased toward simple solutions — and there are far fewer simple solutions to agree on. Two people asked to 'name a number' will almost surely differ; asked to 'name the smallest prime above 100', they almost surely coincide — not through communication, but because the criterion has few satisfiers. Simplicity bias converts 'some solution' into approximately 'the solution', so independent training runs land near it independently. This mechanism also predicts why capability matters: stronger models approach the optimum more closely, and the optimum is where the agreement lives.

The measurement. Because coordinates are meaningless across models (permutation symmetry, Section 2.1), the hypothesis is scored coordinate-free: mutual nearest-neighbour overlap — whether a concept has the same neighbours in model A as in model B, using only within-space distance rankings compared across spaces. Rotation, scaling, permutation and even dimensionality cancel; the metric asks exactly one question — do the two maps share adjacency structure, i.e. shape? It is a coarser cousin of the CKA used in Experiment A (which compares full Gram matrices), and the same signal the unsupervised translation demo operationalizes (Gromov-Wasserstein matches the distance matrices; CSLS refinement lives on mutual nearest neighbours).

The empirical claim and this project's position. The paper's central finding is that alignment *rises with capability*, across model families and modalities. This project samples that curve at specific points: Experiment A is a two-point sample at the low-capability end (two small, independently trained language models — substantial alignment, R-squared 0.74-0.80, against a random-weights zero), Experiment B is its cross-modal engineering instance (a vision-to-vision bridge deployed to production form), and the pair-free translation demo is its strongest form (recovery from shape

alone). The epistemic position, stated precisely: the experiments confirm the measured points and are consistent with the predicted trend; the trend itself — convergence growing with scale toward a shared model of reality — remains an extrapolation that this small-scale replication renders credible but cannot prove, since proof requires measurement along the capability axis, not at one point on it.

Appendix B — Pair-Free Translation (Stage B10): a boundary-mapping negative result

This appendix reports an additional experiment, run after the main study, testing the strongest form of the convergence claim: translation between the two embedding spaces without any paired examples (in the spirit of [vec2vec](#), 2025).

B.1 Design

The project's 4,000 image pairs were split into two disjoint image sets: MobileCLIP embeddings of images 0-1999 on one side, SigLIP embeddings of images 2000-3499 on the other - no image appears on both sides, so nothing links the sets except the shape of the spaces. A simplified, transparent translator was learned unsupervised: Gromov-Wasserstein optimal transport aligning the two intra-space distance matrices (the mathematical formalization of matching shapes without correspondences; [Alvarez-Melis and Jaakkola 2018](#)), followed by a CSLS-corrected mutual-nearest-neighbour self-learning loop refitting a ridge map. Grading used 500 true held-out pairs the training never saw, against a supervised ridge reference, a random-matrix floor, and chance (0.2%). Pre-registered criteria: PASS at $R@1 \geq 10\times$ chance and $\geq 25\%$ of supervised.

B.2 Result

Variant	Mean cosine	R@1	R@5	R@10
Supervised ridge (reference)	0.835	0.910	0.980	0.990
Unsupervised (no pairs ever)	0.395	0.002	0.004	0.006
Random matrix (floor)	0.017	0.000	0.010	0.014

Table B1: Pair-free translation graded on 500 held-out true pairs (chance $R@1 = 0.002$).

The outcome splits cleanly in two. **Distribution-level alignment was recovered from shape alone:** mean cosine to the true targets of 0.395 against a random floor of 0.017 - a 23x effect obtained without a single paired example. **Instance-level correspondence was not recovered:** $R@1$ remained at chance, and the self-learning loop converged to a stable, self-consistent but incorrect matching (428 mutual pseudo-pairs, frozen from iteration 4). The translator finds the right region of the target space but not the right points.

B.3 Bounded diagnostic study

Two hypotheses for the failure were tested and one survived. **(1) Calibration mismatch - refuted.** The conjecture that SigLIP's compressed cosine calibration (Section 4.2) breaks square-loss GW was tested on a synthetic testbed with simulated calibration compression: the method scored $R@1 = 0.92$ regardless, and the proposed normalization fix actively harmed performance. **(2) Manifold structure and anisotropy - supported.** The clustered synthetic testbed on which the method was validated ($R@1 = 144\times$ chance) was rebuilt to resemble real embeddings: a continuous manifold instead of discrete clusters, heavy-tailed density (hubness), and an anisotropy difference between the two spaces - the same property the Procrustes experiment measured between these very models (Section 4.3). On that testbed the method collapsed to chance, and a hyperparameter sweep (transport regularization, rank-normalized distances, larger samples) did not change the outcome. Discrete cluster structure gives shape-matching its anchor points; continuous, differently-stretched manifolds take them away.

B.4 Conclusion

Shape information alone suffices for coarse, distribution-level alignment between independently trained embedding spaces, but instance-level pair-free translation at this scale is beyond linear maps seeded by shape matching: the published vec2vec results employ adversarial training, cycle-consistency, vector-space-preservation losses and nonlinear adapters, and this experiment indicates that machinery is necessary rather than decorative. The finding composes with the main study: the Procrustes experiment showed the two spaces differ in anisotropy beyond rotation, and the diagnostic here shows that same property is plausibly what defeats cheap shape-only matching. This is the project's third boundary-mapping negative result, in the same pattern as the MLP comparison: the failed stronger method is itself the measurement - here, of exactly how much geometry alone can and cannot deliver.

Appendix C — Methodological corrections during the study

Two changes were made to Experiment A after its first run fell short of the pre-registered stitching criterion. Both are recorded here in full, with what changed, why, and what was deliberately left untouched, so that the final numbers can be read with the process that produced them visible.

C.1 What changed

Parameter	Run 1	Run 2 (final)	Reason
Sample count	1,646	10,000	1,646 passages give ~1,234 training rows for a 768x768 fit = 1.6 samples per input dimension. Ridge regression in that regime shrinks hard toward zero and reports systematically pessimistic held-out R2.
WikiText split	validation	train	Mechanical necessity, not a methodological choice: the validation split contains only ~1,646 passages over 100 characters, so 10,000 could not be drawn from it.
Everything else	-	unchanged	Same two models, same random-weights control, same pooling, same max length, same ridge penalty, same 75/25 fit/eval split, same metrics and thresholds.

Table C1: The complete difference between the two runs.

Which ratio governs, and why (clarifying Section 3.2). Section 3.2 states two numbers side by side - a 768x768 matrix of roughly 590,000 parameters, fitted on about 1,234 rows - and the relationship between them deserves to be explicit, because only one of them sets the statistical requirement. Multi-output ridge has the closed form $W = (X'X + \alpha I)^{-1} X'Y$, in which the inverted matrix depends on the inputs alone: the outputs never enter it. The problem therefore decomposes into 768 independent regressions, one per output dimension, each estimating 768 coefficients from the same design matrix and the same shared inverse. They do not interact and do not compete for data. The question that determines whether the fit is well-posed is asked once, per output regression: are there enough rows to estimate 768 coefficients? The governing ratio is therefore **rows per input dimension** - $1,234/768 = 1.6$ in run 1, and $7,500/768 = 9.8$ in run 2, against a conventional minimum of roughly 5 to 10. Rows divided by the 590,000 total parameter count (0.002 and 0.013) is not a difficulty measure: the fit is 768 tractable problems, not one intractable one. A useful consequence is that output width is nearly free, which is why the adapter of Experiment B - mapping 512 dimensions into 768 - needs rows relative to 512, and 3,000 image pairs sufficed there.

C.2 Why using the train split raises no leakage concern here

The train/validation/test convention exists to stop a *model being trained* from being evaluated on data it has already fitted. Each split has a role defined by how much information has leaked into the model: train (full leakage, by design), validation (indirect leakage, through the experimenter's tuning decisions), test (none, if used once). Those roles presuppose a training process. Experiment A has none.

In a normal training pipeline	In Experiment A
The model updates weights on the train split.	GPT-2 and Pythia are frozen. No gradient is ever computed. The text is probe stimuli - inputs used to elicit activations, like a stimulus set in a perception experiment.
The validation split guards hyperparameter choices.	No hyperparameters of the models are chosen; the only fitted objects are the stitching matrices in A3.
The test split gives the honest final number.	A3 holds out its own 25% of the 10,000 samples. Every reported R2 is measured on rows the ridge fit never saw. That is where held-out discipline lives in this study.

Table C2: The split names describe roles in WikiText's intended use (language-model benchmarking), not roles in this experiment.

Both splits are the same corpus, partitioned at the article level from the same set of verified Wikipedia articles and cleaned by the same pipeline; they differ only in which articles landed in which file. The distribution of the property this experiment actually depends on - well-formed English prose - is statistically indistinguishable between them. An analogy: to test whether two thermometers agree, one needs water samples at various temperatures; it is irrelevant which bucket the water came from, because those labels belonged to someone else's experiment.

C.3 A real asymmetry, disclosed

One genuine caveat exists in this area, and it is unrelated to the split choice. WikiText is Wikipedia text. Pythia was trained on The Pile, which *includes* an English Wikipedia component, whereas GPT-2's WebText was constructed with Wikipedia documents removed. Pythia has therefore likely encountered text of this kind during training and GPT-2 has not. Three points bound the consequence: the asymmetry applies identically to the validation and train splits, so it is not affected by the correction described here; the random-weights control processes exactly the same text, so the trained-minus-random gap remains a valid contrast; and the hypothesis under test concerns whether a linear map exists between two representations of the same input, not how surprising that input was to either model. The asymmetry is reported for completeness rather than because it is believed to bias the result.

C.4 Why this is diagnosis and not fitting the result

The distinction that matters: the change was motivated by a *diagnosed confound* identified before the second run, with an explicit prediction attached - if sample starvation was the cause, R2 would rise substantially; if the gap was real, it would plateau. Three pieces of evidence support the diagnosis rather than a lucky reroll. The arithmetic was checkable in advance (590K parameters, 1,234 rows). CKA, which is far less sample-hungry than regression, barely moved between runs (0.365 to 0.359 mean), indicating the underlying similarity was stable and only the regression estimate was distorted. And the control moved in the theoretically predicted direction (-0.773 to +0.173), since with adequate data ridge can exploit residual input correlations even through random layers. The run-1 numbers are reported in full in Section 3.2 rather than discarded.

C.5 One threshold reported as missed

For completeness: the pre-registered CKA criterion was written as 'diagonal CKA mean > 0.5 with random baseline < 0.15'. The final run produced a diagonal mean of 0.424 - below that line - with a random baseline of 0.067 and a trained/random ratio of 6.3x. The absolute threshold was set optimistically: absolute CKA between models with different tokenizers is expected to be modest, and the ratio to the control is the load-bearing test, since it is the quantity that isolates learning from architecture and input statistics. The overall verdict for Experiment A therefore rests on the ratio criteria and the stitching results, all of which passed decisively. This is recorded as a missed sub-threshold rather than a retro-fitted one.

C.6 Direction symmetry, and the anisotropy that masked it

The stitching fit in Section 3.3 maps GPT-2 into Pythia's space. Because the hypothesis is symmetric, the reverse fit should score comparably; a follow-up run tested every layer pair in both directions. Under the variance-weighted R2 used throughout, the reverse direction scored slightly higher in middle layers (mean 0.829 against 0.751) and lower at layer 0 (0.455 against 0.591), for a mean absolute per-layer difference of 0.054. Two further tests identified both effects, and neither reflects a failure of the isomorphism claim.

Layer 0 - a real information difference, architectural in origin. GPT-2 adds learned absolute positional embeddings at the embedding layer; Pythia applies rotary encoding inside attention, so its

layer 0 carries no positional term. Mean pooling therefore leaves a deterministic function of sequence length in GPT-2's pooled layer-0 vector and not in Pythia's. Measured directly: sequence length is recoverable from the pooled layer-0 vector with held-out $R^2 = 1.000$ for GPT-2 and 0.126 for Pythia. Predicting Pythia from GPT-2 is therefore unimpeded, while the reverse requires reconstructing information absent from its input - exactly the observed asymmetry, and the one place in this study where one space demonstrably contains something the other lacks.

Middle layers - an artifact of the scoring metric. Recomputing the same fits with uniform averaging across output dimensions instead of variance weighting collapses the gap: mean absolute difference falls from 0.097 to 0.006, and the two directions become indistinguishable (layer 4: 0.722 forward, 0.723 reverse; layer 5: 0.744 and 0.735; layer 6: 0.748 and 0.740). The cause is measurable anisotropy. Effective dimensionality (participation ratio of the covariance spectrum, out of 768) is far lower for GPT-2 than for Pythia at every depth:

Layer	GPT-2 effective dim	Pythia effective dim
0	8.9	92.7
4	3.3	9.3
8	10.9	19.8
12	2.2	43.9

Table C3: GPT-2's variance concentrates into as few as two or three effective directions - the documented rogue-dimension phenomenon, measured here in the project's own activations. A variance-weighted score with GPT-2 as target is therefore dominated by a handful of easily predicted directions, which inflates the reverse fit.

Conclusion. With the weighting artifact removed, the linear correspondence is symmetric in both directions, as the isomorphism hypothesis predicts, with a single architecturally explained exception at the embedding layer. The forward figures reported in Section 3.3 are essentially unchanged under either weighting (layer 4/5/6: 0.737/0.752/0.754 variance-weighted against 0.722/0.744/0.748 uniform), so the main result is robust to the metric choice. This check also connects the two experiments: the same property that inflates one direction here - unequal variance spread between two models' spaces - is what defeats rotation-only alignment in Experiment B.

C.7 The CKA heat map: what the structure actually is

Section 3.2 describes the trained-versus-trained CKA panel as showing a hot diagonal. Read by eye that is what it suggests; measured, the structure is different from - and in one respect stronger than - that description. This section records the full analysis.

The depth trend, which needs no correction. The diagonal entries rise strictly monotonically across nine consecutive layers, from 0.125 at layer 3 to 0.742 at layer 11 (Spearman $\rho = 1.000$, $p < 10^{-5}$). This is the raw diagonal, with no normalization applied, and it mirrors the stitching curve of Section 3.3 over the same span. Two methodologically independent measures - a coordinate-free similarity index and a fitted linear map evaluated on held-out data - therefore agree that correspondence strengthens with depth. Across all thirteen layers the trend is not significant ($\rho = 0.253$, $p = 0.40$), because the embedding layer and the final layer both fall away sharply; the monotone segment is bounded by exactly the two depths where the architectures differ most.

Layer-to-layer correspondence, which is softer than claimed. Taking each row's argmax, the best-matching Pythia layer lies on average 2.85 layers from the diagonal, and only 62 per cent of layers match within two. The diagonal is nonetheless elevated - 0.425 against an off-diagonal mean of 0.354, a ratio of 1.20 with a 5,000-draw permutation test giving $p < 0.001$. So the diagonal is genuinely above chance without being the sharp ridge the phrase 'hot diagonal' implies.

Why the panel looks blocky: hubness. The apparent block structure is a marginal effect. Row means range from 0.191 to 0.572 - a threefold spread - and a single rank-one component accounts for 97 per cent of the matrix energy, meaning most cells are predicted by 'how similar is this layer to everything' rather than by any specific pairing. Dividing out the row and column marginals (the standard lift correction) sharpens the picture: the diagonal is enriched to 1.16 against an off-diagonal 0.98 ($p = 0.006$), the proximity correlation nearly doubles from +0.131 to +0.242, and the share of layers matching within two rises from 62 to 77 per cent.

Statistic	Raw matrix	Marginals divided out
mean argmax - layer index	2.85	2.46
layers matching within two	62%	77%
diagonal vs off-diagonal	0.425 / 0.354 (1.20x)	1.16 / 0.98 (1.18x)
corr(similarity, proximity)	+0.131	+0.242
permutation p on the diagonal	<0.001	0.006
diagonal depth trend, L3-L11	$\rho = 1.000, p < 10^{-5}$	(unchanged - raw)
trained vs random contrast	5.3x (0.359 / 0.068)	(unchanged)

Table C4: measured structure of the CKA matrix. For calibration, the same statistics on a synthetic matrix with a constructed diagonal give mean offset 0.00, ratio 2.55 and correlation +0.83; on a synthetic block pattern they give 6.00, 1.12 and -0.08.

Consistency with the rest of the study. CKA is invariant to rotation, permutation and isotropic scaling, but not to differences in anisotropy, which ridge regression can absorb and CKA cannot. Section C.6 measured effective dimensionalities of 2.2 to 10.9 for GPT-2 against 9.3 to 92.7 for Pythia. A hubness-dominated CKA matrix with a modest diagonal, sitting beside a stitching R2 of 0.737, is the expected signature of two spaces carrying similar content with very different variance spread - not a disagreement between stages.

Corrected wording for Section 3.2. CKA between the trained models is 5.3 times the random-weights control. The diagonal is elevated (ratio 1.20, permutation $p < 0.001$) and rises strictly monotonically from layer 3 to layer 11 ($\rho = 1.000, p < 10^{-5}$), mirroring the stitching curve. Layer-to-layer correspondence is soft in the raw matrix, which is dominated by hubness; with marginals divided out, 77 per cent of layers match within two and the diagonal is enriched at $p = 0.006$. The phrase 'hot diagonal' overstates the raw panel and should be read as replaced by this section.

C.8 A cross-modal extension: an image-only encoder against a text-only encoder

Sections 3 and 4 compare models within one modality, where row i is literally the same input encoded twice. A follow-up experiment tests the harder claim the Platonic Representation Hypothesis actually makes: whether a vision model trained without any language supervision and a text model trained without any images build representations related by one linear map. Here row i is an image and the captions describing it - different objects sharing a referent.

Design. Image side: DINOv2-base, self-supervised, chosen specifically because it has never received text supervision (any CLIP-family encoder would make the test circular). Text side: bge-m3. Data: 8,000 COCO train2017 image-caption pairs, giving 7,000 training rows against an image width of 768, or 9.1 rows per input dimension - above the threshold whose violation caused the run-1 failure of Section C.1, and checked before the results were read. Fit: ridge, image to text, evaluated on 1,000 held-out pairs. Success bands were fixed before running.

The ceiling must match the task. An early version scored the system against Experiment B's 0.630, which was measured for text-to-image retrieval with one caption per query. Two mismatches were corrected: the direction (E1 runs image to text) and the caption treatment (the system retrieves against

averaged caption embeddings, so the ceiling must too - averaging yields a cleaner target per image and makes retrieval easier). With SigLIP's own towers measured on the same rows under the same treatment, the ceiling is 0.875, not 0.630, and the system's share falls from an inflated 55.2 per cent to 40.9.

Measurement	Result	Reading
shuffle control gap	$0.511 - (-0.130) = 0.641$	three times the 0.2 threshold: the pairing is sound and the fit has power
held-out R2 / cosine	$0.511 / 0.889$	86% of the same-modality vision pair's 0.592, on a target that is an average of five captions
retrieval R@1 / R@5 / R@10	$0.358 / 0.706 / 0.849$	358x, 177x and 71x their respective chance rates
raw cross-space floor	$0.001 / 0.004 / 0.012$	chance - the correspondence exists only through the fitted map
matched ceiling (SigLIP, same rows and treatment)	$0.875 / 0.981 / 0.988$	a jointly trained model, scored on the identical task
share of ceiling	$40.9\% / 72.0\% / 85.9\%$	the share rises steeply with K

Table C5: cross-modal results under the averaged-caption protocol, with a direction- and treatment-matched ceiling. Pre-registered verdict: measurable but weak.

The verdict is robust to the caption protocol. The experiment was run twice, once with a single caption per image and once with all five averaged, each against its own matched ceiling. Both land in the same pre-registered band:

Protocol	R@1	R@5	R@10	verdict band
single caption	24.7%	48.8%	65.6%	measurable but weak
five captions averaged	40.9%	72.0%	85.9%	measurable but weak

Table C6: the obvious objection - that the result depends on how captions were handled - is answered by measuring both. The averaged protocol is the better estimate of what an image is about; the single-caption protocol follows the standard COCO benchmark convention. They agree on the verdict.

Interpretation. The correspondence is real and weaker than within modality. Against Experiment B's same-modality shares of 88.4, 95.1 and 97.2 per cent, the cross-modal figures are 40.9, 72.0 and 85.9: crossing modalities without joint training costs roughly half the rank-1 performance and about a tenth by rank ten. Two features deserve emphasis. First, this is a stricter test than Experiment B, where both encoders had been trained with language supervision and their spaces were therefore already shaped by text; DINOv2 has had none, so whatever correspondence exists arose from two models independently representing the same world with no shared supervisory signal. Second, the share of ceiling rises steeply with K, which is the signature of a map that recovers the correct semantic neighbourhood but not the exact item.

A note that reflects back on Section 4.3. SigLIP scores $R@1 = 0.648$ against single captions and 0.875 against averaged ones on the same images. A substantial part of what Section 4.3 attributes to inherent caption ambiguity is therefore single-annotator phrasing noise rather than irreducible ambiguity. That does not affect the adapter results, which follow the standard one-caption COCO protocol throughout, but the ceiling should be described as protocol-dependent rather than intrinsic.

What this does and does not license. It does not demonstrate that cross-modal representations converge in general; it is one measured point, at one model scale, in one language and one image domain. It does establish that linear cross-modal correspondence is measurable far above chance without joint training, and it converts the limitation stated in Section 5 - that this study samples two points of a hypothesised trend and cannot observe its slope - into a concrete next measurement: repeating the identical protocol with a small, base and large image encoder yields three points against

capacity, which is the variable the hypothesis is actually about. Two cautions apply to that sweep: DINOv2's output width grows with model size, so the number of pairs must grow with it to hold rows per input dimension constant, and the widths should additionally be matched by projection so that a rising line reflects convergence rather than fit capacity. A rising line would be direct evidence for scale-dependent convergence; a flat one would be an equally reportable negative.

C.9 Post-submission experiment series (E1.1, F1-F5): caption protocol, distillation control, refit adapter, metric degeneracy

Why five captions, and what averaging changes. COCO supplies five independent human captions per image; five is the dataset's structure, not a tuned choice. The decision was whether to use one or all. A single caption carries one annotator's emphasis - one names the activity, another the setting - so the target mixes image content with individual phrasing. Averaging the five embeddings cancels the idiosyncrasies and retains what they agree on, making the target 'what the image is about' rather than 'what one person wrote'. The effect is large: held-out R-squared rises from 0.278 to 0.511 and R@1 from 0.160 to 0.358. The same averaging must be applied to the ceiling, since it eases the task for any method - SigLIP itself scores 0.648 on single captions and 0.875 on averaged ones. Scoring an averaged system against a single-caption ceiling produced an inflated 55.2 per cent; matched properly it is 40.9. Both protocols are reported in Table C6 and both fall in the same pre-registered band, so the verdict does not depend on the choice. The standard COCO retrieval convention is one caption per query, so averaging is a deliberate and disclosed deviation, adopted to lower target noise rather than to raise the score.

The MLP control, retested at scale (F1-F2). Section 4 reads the MLP's +0.003 gain as evidence that the MobileCLIP-to-SigLIP relationship is linear. That reading had an untested alternative: the MLP (1.31M parameters) saw only 3,000 examples and, unlike a linear map, does not decompose into independent per-output regressions - it may simply have been data-starved. A 38,000-image teacher corpus (SigLIP 2 targets; no captions required, since the teacher supplies the target) settles it: retrained at 74 rows per input dimension, the MLP's held-out cosine converges to exactly the linear baseline - 0.9052 against 0.9052, gain -0.0000, plateau by epoch 10. The data-starvation reading is refuted; linearity is confirmed by a second, stronger test. The governing ratio differs by architecture: rows per input dimension for the decomposable linear map, examples per parameter (here 0.029) for the MLP - the plateau, not the ratio, is the evidence.

The shipped adapter improves by refitting alone (F5). The shipped W was fitted at 5.9 rows per input dimension - above this report's threshold of 5 but at the bottom of its 5-10 band. Refitting the identical closed-form ridge on the 38,000 corpus rows and evaluating on 5,000 fresh queries across five disjoint 1,000-image galleries (train2017 images beyond the corpus, unseen by both maps) yields significant gains at every rank under a paired bootstrap:

Metric	shipped W	refit W	paired gain	95% CI	verdict
R@1	0.565	0.590	+0.0244	[+0.0162, +0.0330]	real
R@5	0.847	0.858	+0.0116	[+0.0054, +0.0178]	real
R@10	0.918	0.929	+0.0106	[+0.0060, +0.0152]	real

Table C7: refit-adapter evaluation, pooled over 5,000 queries; 300 gained against 178 lost at rank 1. The artifact is unchanged in kind - the same 0.79 MB matrix, the same seconds-long refit on unlabeled images.

The pre-registered R@1 gate of 90 per cent of ceiling, recorded as missed in Section 4.3 at 88.4 per cent, is best described as **approached rather than met**: the refit map reads 90.6 per cent on the original 1,000-query protocol but 89.0 per cent on the expanded evaluation. A smaller-sample point estimate that crossed the line did not survive five times the data - the lesson of Section C.1, applied to evaluation rather than to fitting.

A metric-degeneracy result (E1.1). Replacing the contrastively trained text encoder of Section C.8 with GPT-2 mean-pooled - all else identical, a matched pair - retains 74 per cent of the held-out R-squared (0.377) but only 27 per cent of R@1 (0.095, still 95x chance; shuffle gap 0.505). The mechanism was measured before fitting: GPT-2's caption space has an effective rank of 6.5 of 768 and a mean inter-caption cosine of +0.999, against DINOv2's 380.2 and +0.031. In that space cosine is degenerate - it reads 1.000 for a map whose retrieval runs at 9.5 per cent - the mirror image of Section 4, where R-squared understated a system whose cosine was honest. Together: no similarity metric is universally correct; the geometry of the space determines which metrics are interpretable, and it must be measured first. A label-free rescue test (centering, top-component removal, whitening; notebook E1.2) is built and pending - it separates

'correspondence absent' from 'isotropy absent'.

An initial scale-sweep point (E1, dinov2-large). Section C.8 notes that this study samples a hypothesised trend at one encoder size and cannot observe its slope. A first additional point has been measured: DINOv2-large with cls+patch pooling - the CLS token concatenated with the mean of the patch tokens, DINOv2's own linear-probe protocol, giving a 2,048-dimensional input - fitted on 22,755 rows (11.1 per input dimension) against the same averaged-caption targets. It is the strongest cross-modal result in the project: held-out R-squared 0.609, above the same-modality vision pair's 0.592; cosine 0.912; R@1 0.504, or 58.3 per cent of a matched ceiling of 0.865, against 40.9 per cent for the base encoder; shuffle gap 0.709; raw floor at chance. R@10 reaches 94.2 per cent of ceiling.

This is not yet a capacity measurement. Four factors differed between the two runs - encoder (base to large), pooling (cls to cls+patch), input width (768 to 2,048) and sample count (8,000 to 23,755) - so the 17.4-point gain cannot be attributed to encoder capacity. Sample count is a deliberate control, scaled to hold rows per input dimension near 10, and width is largely neutralised by that same scaling, since rows per parameter then stays constant. Pooling is the substantive confound: cls+patch changes what information the vector carries, not merely how many dimensions hold it, and is known to improve linear probes independently of model size. A matched slope therefore requires the smaller encoders re-run under cls+patch, which is in progress; MATCH_DIM offers an additional capacity-controlled variant.

Note on regularization. ALPHA_GRID selects ridge regularization strength on a validation slice carved from the training rows; it is a search, not a learning-rate schedule, since the ridge solution is closed-form. Its sensitivity is itself diagnostic: in the dinov2-large run every candidate scored 0.596, the problem being well-conditioned at 22,755 rows against 2,048 dimensions, whereas in E1.1 an alpha of 10 produced R-squared -0.741 in GPT-2's near-degenerate space.

Training status across the project. Experiments A, B and E fit only closed-form matrices under no_grad; every pretrained model remains bit-identical to its download. The F series is the deliberate contrast condition: F2 trains a separate head beside the frozen encoders; F3, the only notebook that would modify encoder weights, is built and deliberately unrun.

C.10 The scale sweep: cross-modal alignment against image-encoder capacity

Section 5 lists as a limitation that this study samples two points of a hypothesised trend and cannot observe its slope. The Platonic Representation Hypothesis is fundamentally a claim about that slope - that representational alignment grows with model capability - so a two-point sample is consistent with it but cannot test it. This section replaces that limitation with a measurement.

Design. The cross-modal protocol of Section C.8 was repeated with three image encoders of the same family and training procedure - DINOv2 small, base and large, 22M to 304M parameters, a 13.8-fold range - holding every other factor fixed: the same text encoder (bge-m3), the same COCO images and five-caption averaging, the same cls-plus-patch pooling, the same ridge protocol with a validation alpha sweep, and the same direction- and treatment-matched SigLIP ceiling measured per run. Two controls make the comparison a capacity measurement rather than a confound. First, pooling is held constant: an earlier base run used CLS-only pooling and is excluded, because concatenating the patch mean changes what information the vector carries and is known to help linear probes independently of model size. Second, the number of pairs scales with the encoder's output width so that rows per input dimension is 11.1 at every point - without this, wider encoders would be progressively starved, and starvation biases downward, which would understate exactly the trend under test.

Encoder	Para ms	Width	Pairs	rows/di m	R2	shuffle gap	R@1	R@5	R@10	% ceiling @1 / @5 / @10
DINOv2-small	22M	768	9,533	11.1	0.516	0.616	0.357	0.698	0.826	40.9 / 71.3 / 83.4
DINOv2-base	87M	1,536	18,066	11.1	0.580	0.682	0.466	0.809	0.907	53.5 / 82.2 / 91.2
DINOv2-large	304M	2,048	23,755	11.1	0.609	0.709	0.504	0.835	0.935	58.3 / 85.6 / 94.2

Table C8: the sweep. Width is the pooled output (CLS concatenated with the patch mean, hence twice the hidden size). Ceilings were measured per run on that run's own held-out rows and fell between 0.865 and 0.872 at R@1, so the percentage column is comparable across rows.

Result. Every measure rises monotonically with capacity: held-out R-squared, mean cosine, all three recall levels, and the shuffle-control gap. Six measures moving together across a 13.8-fold parameter range is a considerably stronger signal than any one of them alone, and the rising shuffle gap is worth noting separately - it shows the fit is pulling more signal relative to its own misaligned control, not merely scoring higher. Against the logarithm of parameter count the trend is close to linear: R@1 gains 15.3 points of ceiling per decade (fit R-squared 0.949) and held-out R-squared gains 0.082 per decade (0.966). Both base and large exceed this study's same-modality reference of 0.592 from Section 4, on a task that crosses modalities without joint training.

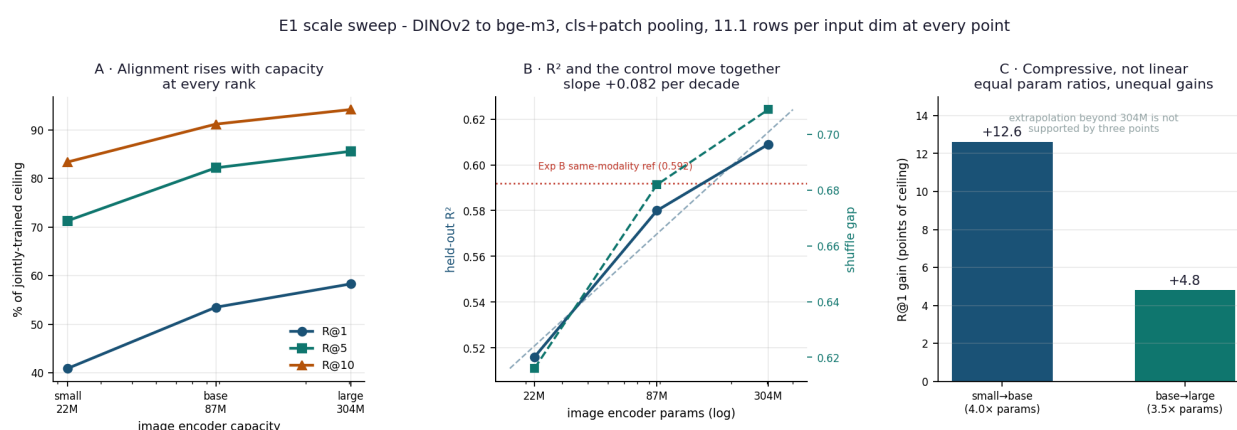


Figure C4: alignment against capacity. Panel A, share of a jointly-trained ceiling at each rank. Panel B, held-out R-squared with the shuffle control on the secondary axis, both against log parameters. Panel C, the gains from equal parameter ratios.

The returns are compressive. Roughly equal parameter ratios do not buy equal improvement: small to base is a 4.0-fold increase and gains 12.6 points of R@1 ceiling share, while base to large is 3.5-fold and gains 4.8. The log-linear fit is therefore a convenient summary of three points rather than a law, and this section makes no claim about behaviour beyond 304M parameters. A flattening curve is what one would expect as the ceiling is approached - large already reaches 94.2 per cent of it at rank ten - but three points cannot distinguish

saturation from a genuinely diminishing exponent.

What this establishes. Within the range measured, representational alignment between a vision encoder that has received no language supervision and a text encoder that has never seen an image increases with the vision encoder's capacity. That is the Platonic Representation Hypothesis's central prediction, tested in the setting where it is least trivially true, with pooling and statistical power held constant across points. It remains one model family, one text encoder, one image domain and one language, and the appropriate reading is a measured trend within those bounds rather than a general law. The limitation stated in Section 5 - that the slope could not be observed - is discharged; the narrower limitation that replaces it is that the slope has been observed over 13.8-fold, in one family, and is compressive at the top of that range.

C.11 Isotropy, not similarity training: a label-free correction recovers the collapsed retrieval

Section C.9 reported that substituting GPT-2 mean-pooled for the contrastively trained text encoder retains most of the held-out R-squared but collapses retrieval, and concluded that usable retrieval geometry requires similarity training. That conclusion conflated two claims it never separated: that the correspondence is absent from raw language-model representations, and that the coordinate frame merely hides it. This section separates them, and the second is correct.

Design. Four corrections were applied to GPT-2's caption space and the identical fit-and-evaluate protocol re-run under each. All are purely geometric and label-free - no contrastive objective, no pairs, no gradient - and every transform is estimated on the training targets only and applied unchanged to the held-out rows. Image side: DINOv2-base with cls-plus-patch pooling, 17,066 training rows at 11.1 per input dimension.

Correction	eff. rank (of 768)	R2	R@1	% of matched bge-m3 arm
DINOv2-base (bge-m3 arm: R@1 0.466)				
raw (C.9 baseline)	6.5	0.440	0.130	27.9%
centered	6.5	0.440	0.104	22.3%
all-but-top-1	18.1	0.402	0.156	33.5%
all-but-top-3	34.7	0.464	0.227	48.7%
all-but-top-10	99.8	0.365	0.320	68.7%
whitened	490.8	0.014	0.478	102.6%
DINOv2-large (bge-m3 arm: R@1 0.504)				
raw	6.4	0.439	0.136	27.0%
centered	6.4	0.439	0.131	26.0%
all-but-top-1	18.3	0.416	0.164	32.5%
all-but-top-3	35.2	0.465	0.252	50.0%
all-but-top-10	101.8	0.386	0.374	74.2%
whitened	495.2	0.034	0.511	101.4%

Table C9: label-free corrections on GPT-2's caption space, at two image-encoder scales. Percentages are against the bge-m3 arm measured with the SAME encoder and pooling - drafts that compared against a different configuration overstated the recovery in both cases, which is why the notebook now derives the reference arm from the pairs filename rather than accepting it as a setting.

Result, replicated at two scales. Whitening the target space - a linear, unsupervised rescaling estimated from training targets alone - lifts R@1 from 0.130 to 0.478 with DINOv2-base and from 0.136 to 0.511 with DINOv2-large: lift factors of 3.7 and 3.8, reaching 102.6 and 101.4 per cent of an encoder contrastively trained on 1.2 billion text pairs for exactly this purpose. The rank correlation between effective rank and R@1 is +0.94 at both scales. The correspondence was present in the raw language-model representations throughout; only the coordinate frame concealed it. The claim that similarity training is required does not survive: what similarity training supplies here is isotropy, and isotropy can be supplied afterwards, for free.

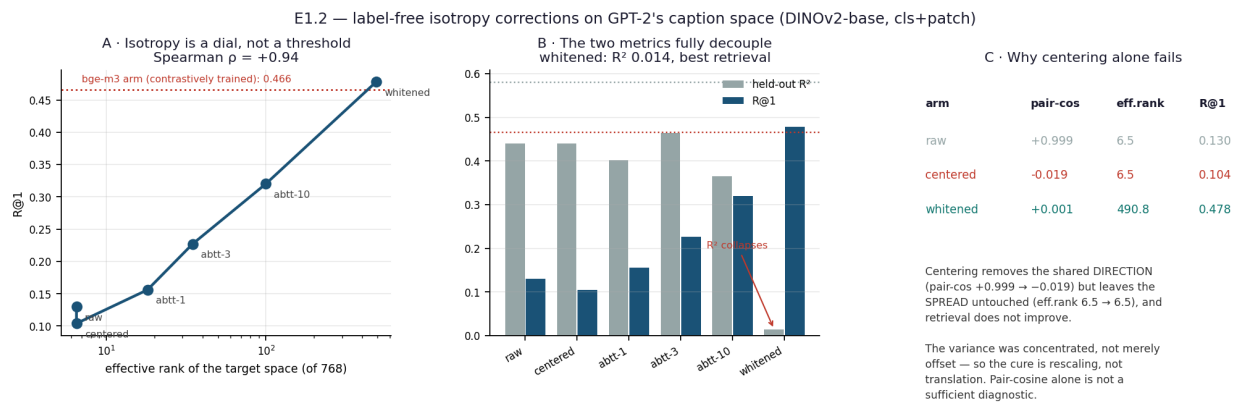


Figure C5: isotropy against retrieval (A), the divergence of R-squared and recall (B), and why centering alone is insufficient (C).

Isotropy behaves as a continuous dial. Across the six arms, effective rank rises 6.5, 6.5, 18.1, 34.7, 99.8, 490.8 and R@1 rises 0.130, 0.104, 0.156, 0.227, 0.320, 0.478 - a rank correlation of +0.94. Retrieval quality is not gated by crossing an isotropy threshold; it tracks the degree of isotropy monotonically.

Centering alone fails, and the failure is instructive. Centering drives mean pair-cosine from +0.999 to -0.019 - the standard symptom of anisotropy, apparently cured - while leaving effective rank unchanged at 6.5, and retrieval does not improve (0.130 to 0.104). The variance was concentrated across directions, not merely displaced by a common offset, so the remedy had to be rescaling rather than translation. Mean pair-cosine is therefore not a sufficient diagnostic on its own; it must be read alongside a spectral measure.

The metrics separate completely. The whitened arm records a held-out R-squared of 0.014 - indistinguishable from zero - while delivering the best retrieval of any arm and beating the contrastively trained reference. Whitening rescales the target space, so predicting it in squared-error terms becomes nearly impossible, while the angular structure that retrieval consumes becomes nearly ideal. Taken with Section 4, where R-squared of 0.592 understated a system whose cosine was 0.900, and Section C.9, where cosine of 1.000 overstated a system retrieving at 9.5 per cent, this completes the argument: a similarity metric is interpretable only relative to the geometry of the space it is computed in, and that geometry must be measured first.

This resolves an apparent contradiction with Section C.10. Measured raw, the GPT-2 arm stops responding to image capacity: moving from DINOv2-base to DINOv2-large gains 0.002 in R-squared and nothing in R@1 (0.160 to 0.138, a difference of 22 queries in 1,000 and within noise), while the bge-m3 arm over the same two encoders gains cleanly at every rank. That looked like the extra capacity being wasted against a degenerate target. It was not: after whitening, the large encoder does benefit - 0.478 to 0.511 - tracking the bge-m3 arm's own base-to-large gain. The additional information was arriving throughout; the coordinate frame was discarding it. The scaling trend of Section C.10 therefore holds against a language-model target as well, but is invisible until the frame is corrected - which is itself the strongest form of this section's argument.

Implication beyond this project. Cross-model alignment is routinely quantified with cosine- and kernel-based metrics. Where either space is strongly anisotropic, those metrics can report an absence of alignment that a three-line unsupervised transform removes. Reported failures of convergence should therefore be checked against an isotropy battery before being believed - the measurement may be describing the coordinate frame rather than the content.

C.12 Recovering the correspondence without any paired examples

Every experiment to this point is given the pairing: row i of one array is the same item as row i of the other, and only the map is fitted. This section removes that last piece of supervision. Two sets of 1,500 vectors are taken from the same images, one side is shuffled, the true permutation is sealed away and used only for final scoring, and the correspondence itself must be recovered from the shapes of the two point clouds. Stage one matches structure to structure with entropic Gromov-Wasserstein - each space is described only by its own internal cosine-distance matrix, so no cross-space pair is ever seen. Stage two fits a ridge map from the assignment that stage one proposes.

Result. Exact matching reaches **39.7 per cent** against a chance rate of 0.067 per cent - 595 times chance - with 53.5 per cent of true partners in the top five and 58.0 in the top ten. The map fitted from that assignment reaches $R@1$ 0.252 and $R@10$ 0.567 in the standard retrieval protocol, which is 40 per cent of the jointly-trained ceiling and 45 per cent of the supervised adapter, achieved with no paired examples whatsoever.

Map	paired examples used	R@1	R@5	R@10	verification AUC
ceiling: SigLIP native	-	0.630	0.885	0.944	1.000
supervised W	3,000	0.557	0.842	0.918	0.999
unsupervised W	0	0.252	0.477	0.567	0.906

Table C10: the unsupervised map against the supervised one and the ceiling, on the same held-out rows. Verification AUC asks whether a threshold separates a true pair from a random one - a weaker question than outranking 999 impostors, and the unsupervised map answers it far better than its retrieval figures suggest.

The entropic regularizer dominates the outcome. An earlier run of this experiment reached only 8.6 per cent exact matching and produced a map at 6 per cent of ceiling. The difference was one hyperparameter: that run used $\epsilon = 5e-4$ with 200 iterations, under which Sinkhorn did not converge. Too small a regularizer makes each projection nearly a hard assignment, which both converges slowly and locks in early mistakes.

epsilon	exact	top-5	top-10	x chance	seconds
1e-3	39.7%	53.5%	58.0%	595x	273
3e-3	12.2%	26.5%	32.6%	183x	28
5e-3	6.6%	16.2%	22.5%	99x	18
1e-2	1.3%	4.6%	8.1%	19x	15

Table C11: the sweep at $N = 1,500$. The optimum is sharp and it moves with N - a preliminary sweep at $N = 800$ placed the best value at $5e-3$, which at $N = 1,500$ yields 6.6 per cent. The regularizer must be tuned together with the problem size rather than carried across settings, which is why the notebook sweeps rather than fixes it.

A deliberate negative: whitening destroys the recovery. Section C.11 established that whitening the target space rescues cosine retrieval, lifting $R@1$ by a factor of 3.7. Applying the same transform to both spaces before Gromov-Wasserstein collapses matching from 39.7 per cent to 0.1 per cent, against a chance rate of 0.067 - complete destruction of the signal, confirmed at two problem sizes. The effective ranks confirm the transform did what it claims: 135 and 171 before, 494 and 743 after.

This is the sharpest distinction the project produces, and it is not a contradiction. Cosine retrieval reads a correspondence that is already known; there, variance concentrated in a few directions is noise that drowns the distinctions retrieval needs, and equalising it helps. Gromov-Wasserstein discovers a correspondence that is unknown, by matching one space's distance structure against the other's; there, uneven variance is precisely the signal, because it is what makes a point's position distinctive relative to its neighbours. Whitening equalises every direction and so erases exactly the structure the matcher depends on. Isotropy helps when reading a correspondence and hurts when finding one - the same geometric property, opposite prescriptions, both measured rather than assumed.

A second negative: iterative refinement degraded the solution. Alternating between fitting a map and re-matching - the standard ICP refinement - reduced accuracy monotonically across ten iterations (8.6, 8.3, 7.9, and so on to 6.9 per cent) and converged stably at the degraded answer. Fitting a map from mostly incorrect

assignments teaches the incorrect correspondence, which is then re-matched onto with greater confidence; convergence there indicates agreement with its own errors rather than correctness. The result reported above is Gromov-Wasserstein alone.

What this establishes. The correspondence between two independently trained encoders is partially self-identifying: it can be recovered from geometry alone, at 595 times chance, without a single paired example. That is the most demanding form of this project's thesis - not merely that the spaces are related by a linear map, but that the relation is discoverable without being told. It remains far weaker than the supervised map and is evidence about the geometry rather than a deployable component; the honest summary is substantial partial recovery, with the mechanism of both its success and its two failures measured.

C.13 The shared space as an object: one hub, five encoders, zero-shot component transfer

Every preceding result is pairwise: encoder A relates to encoder B through a map fitted for that pair. N encoders would need N-squared maps, and nothing in a collection of bilateral fits shows that the encoders share ONE coordinate system. This section constructs that coordinate system and tests it by use. Five cached spaces over the same 9,533 images - DINOv2 small, base and large (no language supervision), bge-m3 captions (contrastively trained), and GPT-2 captions (next-token prediction only) - are each given a single linear map into a 512-dimensional hub: a whitened PCA basis of all five spaces concatenated after per-space standardisation, estimated on 8,533 training rows. Row alignment across caches is verified numerically before anything is fitted, and 1,000 rows are held out for every measurement.

The constructive test. A caption-retrieval head (hub to bge-m3 space) is trained on the hub vectors of ONE encoder - DINOv2-small - and then applied, unchanged, to the hub vectors of encoders it has never seen in any form. A random map into the hub replaces the fitted one as a control, and must sit at chance for the test to mean anything.

encoder	R@1	R@5	R@10	% of native head	
img_small	0.398	0.739	0.875	111.5%	trained here
img_base	0.442	0.792	0.902	95.9%	never seen
img_large	0.468	0.808	0.914	92.9%	never seen
random-map control	0.000	-	-	-	chance = 0.001

Table C12: zero-shot transfer of a single trained component. 'Native head' is a head fitted directly on that encoder's own hub vectors - the upper reference for each row.

Result. The transferred head reaches 95.9 and 92.9 per cent of the native heads on two encoders it never saw. Absolute performance RISES on the unseen encoders - 0.398, 0.442, 0.468 - so the component is not merely surviving transfer: a stronger encoder still helps, through the shared coordinates, using a head fitted on a weaker one. The control sits at chance. A working system has been assembled from parts that were never trained together, which is the Platonic Representation Hypothesis's constructive core: the shared representation exhibited as an object and used, rather than inferred from correlations.

The operating point was measured, not chosen. Hub width and ridge strength were swept jointly, with the project's minimum of five training rows per fitted dimension enforced:

HUB_DIM	rows/dim	a=0.01	a=0.1	a=1.0	a=10
128	66.7	0.390	0.390	0.390	0.390
256	33.3	0.430	0.430	0.430	0.428
512	16.7	0.442	0.442	0.445	0.445
768	11.1	0.006	0.013	0.092	0.341
1024	8.3	0.007	0.009	0.024	0.137

Table C13: transfer R@1 (img_small-trained head on unseen img_base) across hub width and ridge alpha. Verification AUC at the chosen point is 0.979-0.984.

There is a cliff between 512 and 768 dimensions, and it is not a statistical-power artifact - rows per dimension remains above the threshold on both sides. The hub basis is a whitened PCA, which divides each component by its singular value; past the cliff the basis amplifies low-variance directions whose content is noise, and those directions then dominate the cosine geometry that transfer depends on. Below the cliff, alpha is irrelevant across three orders of magnitude; past it, heavy regularisation claws back part of the loss (0.006 to 0.341 at 768) but never reaches the narrow hub's level. Width selection matters; regularisation cannot substitute for it.

Transitivity, and an honest trade-off. Composition through the hub (A to hub to B, with A-to-B never fitted) retains 83 to 95 per cent of directly-fitted maps for every image-space target - essentially meeting the pre-registered 85 per cent bar - and fails into the two text spaces, whose reconstruction the 512-d hub does not

carry (their reach-into-hub R-squared, 0.24 and 0.19, predicts exactly this). Widening the hub repairs reconstruction - retention into the widest targets reached 84 to 88 per cent at 2048 in a separate measurement - but that same widening is what destroys transfer. The two demands pull the width in opposite directions, and no setting satisfies both. A hub tuned for portability is not a lossless reconstruction medium, and vice versa: this is the reading-versus-representing tension of Sections C.11 and C.12 in a third form, and it appears to be a property of shared linear coordinates as such, not of any one experiment.

Scope. Five spaces from two model families, one image domain, one language. What is exhibited is a shared linear coordinate system for these five encoders, demonstrated by component transfer at 93 to 96 per cent of native performance - not a universal space for all models. Within that scope, the central objection to convergence claims - that pairwise fits might each be learning their own idiosyncratic bridge - is answered: one basis serves all five, and work done in it moves between encoders that never met.

C.14 The encoder ladder: Section C.11 as a pre-registered prediction

Section C.11 explains the GPT-2 arm's collapse and rescue with one mechanism: what similarity training supplies is isotropy, not correspondence. An explanation fitted to the data that motivated it is weak evidence; the same explanation predicting the outcome for encoders it has never seen is strong evidence. Two such encoders were therefore run, with the predictions registered in the notebook before the numbers existed: BERT (masked-language modelling, bidirectional - predicted to land BETWEEN GPT-2 and the contrastive encoders in both effective rank and raw retrieval) and Sentence-BERT all-mpnet-base-v2 (contrastive - predicted near the bge arm, with little to gain from whitening). Image side, protocol, caption averaging, and the whitening recipe are identical to Sections C.9 and C.11: the cached DINOv2-base cls+patch embeddings, 18,066 pairs at 11.1 rows per input dimension, transforms fitted on training targets only.

encoder	training objective	eff. rank (of 768)	mean pair-cos	raw R@1	whitened R@1
GPT-2 (C.9/C.11)	causal LM	6.5	+0.999	0.160	0.478
BERT	masked LM, bidirectional	65.0	+0.810	0.362	0.521
SBERT	MLM + contrastive	93.4	+0.139	0.510	0.588
bge-m3 (reference)	heavy contrastive	high	low	0.466	-

Table C14: the ladder. All rows are targets for the same image encoder under the same protocol; only the text encoder's training objective varies.

Prediction 1 - confirmed exactly. Raw retrieval orders by the target space's effective rank with rank correlation **+1.00** across the ladder: 6.5, 65.0, 93.4 in effective rank maps onto 0.160, 0.362, 0.510 in R@1. BERT landed precisely where registered - between GPT-2 and the contrastive encoders - and the severity ordering of the anisotropy itself came out as predicted (pair-cosine +0.999, +0.810, +0.139: degenerate, anisotropic, near-isotropic). Raw cross-modal retrieval against a language-model target is governed by the target's geometry, not by the amount of language understanding its objective implies.

Prediction 2 - partial, and the miss is the sharper finding. Whitening was predicted to collapse the ladder onto one level. It lifted every arm - 0.478, 0.521, 0.588 - but a spread of 0.110 remains, and the residual ordering is IDENTICAL to the raw ordering. The refined statement replaces C.11's original one: **isotropy is necessary but not sufficient**. Whitening equalises how many directions carry variance - all three targets reach effective rank near 765 afterwards - but it cannot install WHICH directions carry meaning. Contrastive training contributes two separable things: a coordinate frame (isotropy), which a label-free linear transform supplies for free and which accounts for most of the gap, and a semantic arrangement of directions, which it cannot. That residue is why a purpose-built contrastive encoder still leads after every arm is whitened.

What this does to the project's claims. C.11's mechanism survives its first out-of-sample test and gains a boundary. The quantitative split is worth stating: for GPT-2, whitening closes the distance from 34 per cent of the bge arm to 103 per cent - the frame was nearly the whole story for the most degenerate space. For BERT and SBERT the lift is smaller and a training-objective residue remains visible. Reported failures of cross-model alignment should still be checked against an isotropy battery first - but after the battery, what remains is a genuine signature of the training objective, and the ladder measures its size.

Appendix D — A derived capability: cross-tier membership and similarity

This appendix extends the engineering discussion of Section 4.5 with a capability the shared index provides at no additional cost, identified after the main write-up.

Because the adapter places phone-side vectors in SigLIP's coordinates, the shared index supports **image-as-query** operations across tiers, not only text search. Computing $\hat{u} = L2(v \cdot W)$ for a photo on the device and querying the index with $\hat{u}$ itself yields, from the top of the ranking: **(a) membership and near-duplicate detection** — if the same photo, or a crop or re-save of it, is indexed on the server, its native SigLIP vector returns with a cosine far above the background, and thresholding on the gap (the triage rule of the B5 sandbox) answers 'is this already in the library?' without uploading the photo; and **(b) cross-tier similar-image search** — lower cosines with a clear ranking give 'server photos like this one', and symmetrically for server photos against phone-indexed content.

Two boundaries keep the claim precise. First, this predicts an image's *address* in SigLIP's space, never its content: W maps 512 numbers to 768 and cannot reconstruct pixels (though, per the security discussion in Appendix B, embeddings should still be treated as content-bearing, since dedicated inversion models recover approximate content from them). Second, the capability exists only through the adapter — a raw MobileCLIP vector compared against the server gallery scores exact chance (Section 4.3), so cross-tier membership is a property the matrix creates, not one the encoders share natively.

The expected accuracy is bounded below by the measured text-query results (Section 4.3), and image-to-image retrieval within one contrastive space is typically easier than text-to-image, since no modality gap is crossed. The measurement is a three-line variant of stage B3 — querying the SigLIP gallery with the adapted vectors instead of the caption embeddings — and is included in the B3 notebook as an optional cell.

D.2 Three decision problems on the shared index, and where a threshold belongs

Appendix D reports the membership capability with a per-query statistic: the true twin outranks the best impostor for 98.7 per cent of queries. That number is correct for what it measures, but 'membership' can be read as a deployable detector - a system that answers yes or no from a score against a cutoff - and that is a different, harder question. This section separates three decision problems that the same adapter and the same 1,000 held-out vectors support, and measures each with the instrument appropriate to it.

Problem	Form	Question	Instrument	Result
Verification	1:1	are these two vectors the same photo?	ROC / AUC	0.999
Ranking	1:N, per query	does the twin outrank every impostor for THIS query?	positive-gap rate	0.987
Membership	1:N, one global cutoff	does a single threshold separate indexed from absent?	ROC / AUC	0.889

Table D2: the same 0.79 MB adapter measured as three different deciders. Raw MobileCLIP without the adapter scores AUC 0.510 on membership and 0.477 on verification - chance - so the entire capability in every row is created by W.

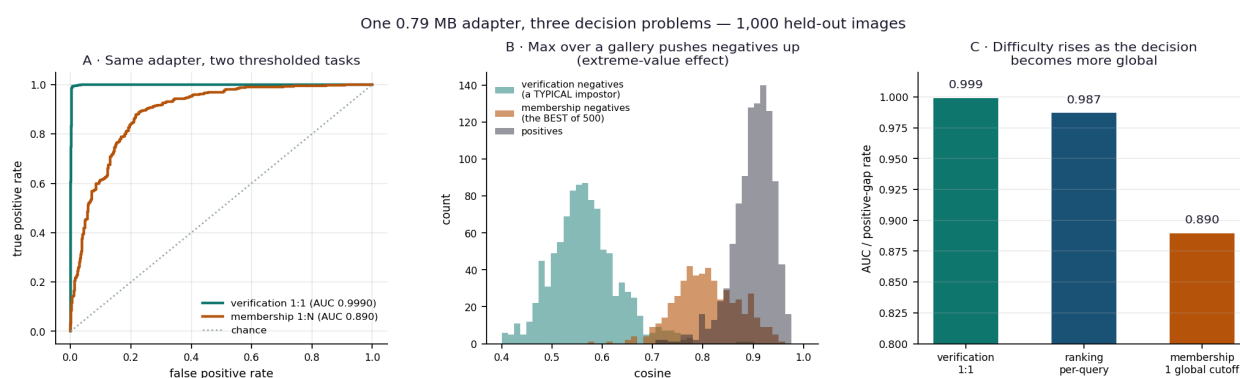


Figure D2: the two ROC curves, the score distributions that explain their difference, and the difficulty ordering.

Why difficulty rises left to right. A verification negative is a typical impostor, and typical impostors score around 0.57 - far below the positives at 0.90, so a threshold separates them almost perfectly. Membership must instead reject the BEST impostor among 500: the maximum of 500 draws sits far into the tail, averaging 0.809, and the distributions overlap. Ranking sits between the two because it also faces the best impostor but only per query - each query needs its twin to win locally, not to clear a cutoff shared with every other query. Cosine scales vary across queries, so one global threshold cannot be optimal for all of them.

Operating points for the deployed system. At a false-positive rate of 1 per cent the membership detector recovers 14.2 per cent of true members; at 5 per cent, 45.0; at 10 per cent, 61.4. The best F1 is 0.840 at a cosine cutoff of 0.856. The honest deployment statement is therefore: cross-tier ranking is near-ceiling, verification of a claimed match is near-perfect, and duplicate DETECTION with one global threshold is good but materially harder - a consumer should prefer per-query comparisons wherever the application allows.

A note on scope. ROC applies to these tasks because each has a binary label and a score. It does not apply to R@K retrieval, which has one correct answer in a gallery and no threshold - the cumulative rank curve is the analogous instrument there. One further observation ties this appendix to Section C.11: in caption verification through the adapter chain, positive cosines average just 0.136 against negatives at -0.024, yet AUC is 0.997 - absolute similarity values are small while the decision is nearly perfect, because AUC measures separation, not magnitude. Which metric is informative depends, again, on the question and the geometry, not on the size of the numbers.

D.3 The allegory, measured

The hypothesis this project tests carries a two-thousand-year-old picture with it. In Plato's cave, prisoners chained before a wall take the shadows cast on it for reality; only a prisoner who is turned sees that the shadows were of solid forms all along. The Platonic Representation Hypothesis borrows the name deliberately: independently trained models are the prisoners, their raw vector spaces the shadows, and the shared structure of the world the forms that both are shadows of. This project's adapter is the turning - not new sight, and not a new model, but a single linear map that rotates one encoder's shadow until it falls on the other's, revealing that the two were casts of one reality.

The figure makes the allegory quantitative. Read as a decision - can a threshold tell a true correspondence from a false one - the three prisoners separate cleanly. Chained (raw MobileCLIP, no adapter) the shadows carry no recoverable signal and the curve sits on the diagonal, AUC 0.481. Turned (the 0.79 MB adapter) verification reaches AUC 0.999: the two encoders name the same forms almost perfectly. Membership under a single global cutoff - the harder act of reading every shadow by one standard - reaches 0.889, the middle curve. The turning is cheap, linear, and touches no pretrained weight; what it reveals is that the agreement was always there to be found.

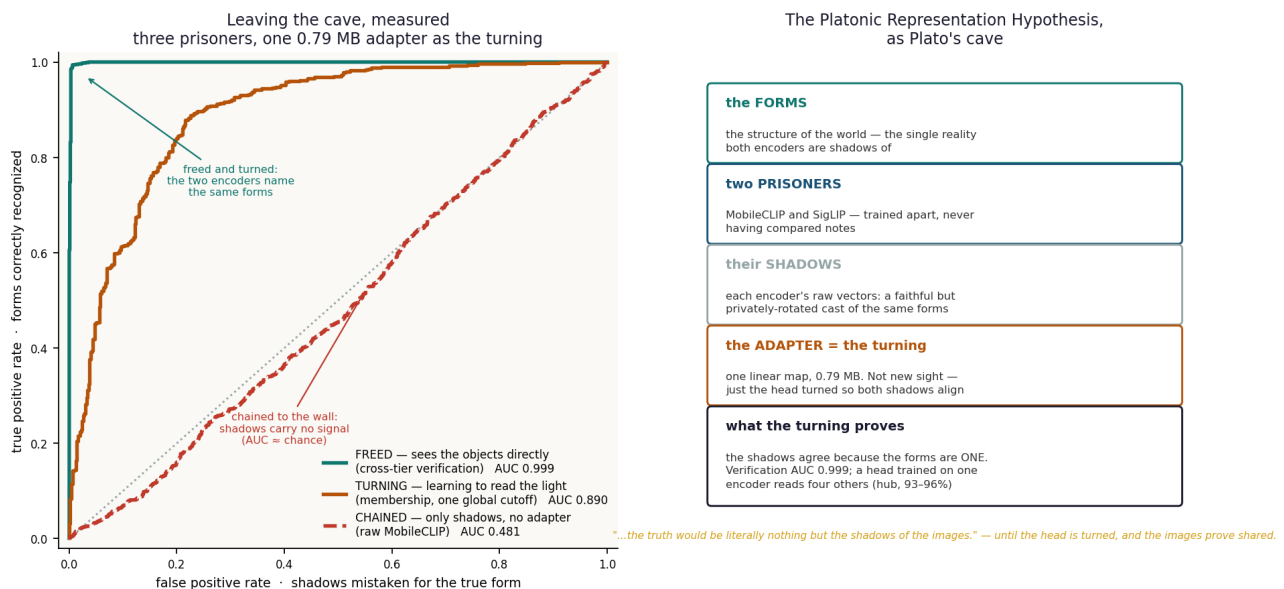


Figure D3: the cave allegory as an ROC. Left, three prisoners as three decision problems on real held-out data. Right, the correspondence between the allegory and the measured system.